Quantum Gravitational Dynamics

Chapter 10: Formal Quantization of QGD

The Complete Master Metric, Path Integral,

Lee–Wick Ghost, Wheeler–DeWitt, and Open Problems

Romeo Matshaba

UNISA DOI:10.5281/zenodo.18827993

March 2026

Abstract

We carry out the formal quantization of QGD starting from the correct complete master metric (including the geometric scaling matrix $M_{\alpha\beta}$). We show that $M_{\alpha\beta}$ plays four distinct roles: encoding gauge choice, the cosmological scale factor, multi-source weighting, and anisotropic deformation. The QGD propagator $\Delta_{\mathrm{QGD}}(k) = 1/k^2 - 1/(k^2 + m_Q^2)$ is a Lee–Wick structure rendering loop integrals UV-finite. The Wheeler–DeWitt quantum bounce at $\sigma_t^* = 1$ is derived from the QGD master metric directly. We identify $\xi_A = 5201\pi^2/656$ as the renormalized self-energy coefficient of the dressed propagator, connecting the Asymptotic Merger Theorem to the gap equation. We catalogue and correct six incorrect claims from circulating analyses.

Contents

1 The Complete QGD Master Metric **3**

1.1 The corrected formula . . . 3

1.2 Signature conventions . . . 3

1.3 The four roles of $M_{\alpha\beta}$ . . . 3

2 The QGD Action and Path Integral **4**

3 Propagator, Feynman Rules, and UV Finiteness **4**

4 The Dyson–Schwinger Gap Equation and ξ_A **5**

5 Wheeler–DeWitt Equation and Quantum Bounce **6**

5.1 ADM decomposition . . . 6

6 Renormalization Group **6**

7 Assessment of Circulating Claims **7**

8 Consequences of the QGD Wheeler–DeWitt Equation **7**
8.1 Information paradox resolution 7
8.2 A dynamical mechanism for small cosmological constant 8
8.3 Big Bounce and inflation from one equation 8
8.4 Black hole entropy correction 8
8.5 Planck-mass remnants as quantum solitons 8
8.6 Modified Unruh effect: three modes 9
8.7 Ladder B from the self-energy cross-term 9
8.8 NLO spin-orbit 9

9 Complete Status Table **10**

1 The Complete QGD Master Metric

1.1 The corrected formula

The complete master metric, incorporating the geometric scaling matrix $M_{\alpha\beta}$, is:

$$\boxed{g_{\mu\nu}(x) = T^\alpha_\mu T^\beta_\nu \left(M_{\alpha\beta} \circ \left[\eta_{\alpha\beta} - \sum_{a=1}^{N} \varepsilon_a \, \sigma^{(a)}_\alpha \sigma^{(a)}_\beta - \kappa \ell_Q^2 \, \partial_\alpha \sigma^\gamma \partial_\beta \sigma_\gamma \right] \right)} \tag{1}$$

with components:

$$\begin{aligned}
T^\alpha_\mu &: \text{ coordinate transformation matrix} \\
M_{\alpha\beta} &: \text{ geometric scaling matrix } (\circ = \text{Hadamard product}) \\
\varepsilon_a \in \{+1, -1\} &: \text{ source signature} \\
\ell_Q = \sqrt{G\hbar^2/c^4} &: \text{ quantum gravitational length} \\
\kappa \approx 2 &: \text{ stiffness coefficient}
\end{aligned}$$

The operation $(M \circ B)_{\alpha\beta} = M_{\alpha\beta} \cdot B_{\alpha\beta}$ (element-wise, no summation). The full Lorentz contraction is carried by $T^\alpha_\mu T^\beta_\nu$.

1.2 Signature conventions

Checking the (t,t) component for Schwarzschild ($\sigma_t = \sqrt{r_s/r}$, $M_{tt} = 1$, $T^t_t = 1$):

$$g_{tt} = \eta_{tt} - \varepsilon_t \sigma_t^2 = -1 - \varepsilon_t \frac{r_s}{r} \xrightarrow{!} -\left(1 - \frac{r_s}{r}\right). \tag{2}$$

This requires $\varepsilon_t = -1$. In QGD: the time σ-field has repulsive signature in the master metric formula, which corresponds to *attractive gravity* (it reduces $|g_{tt}|$ below the Minkowski value). Spatial components have $\varepsilon_r = +1$.

1.3 The four roles of $M_{\alpha\beta}$

Theorem 1.1 ($M_{\alpha\beta}$ resolves four structural questions)**.** *The matrix $M_{\alpha\beta}$ is not a free function but is determined by the specific solution ansatz and plays four distinct roles:*

1. ***Gauge choice.*** *Isotropic gauge: $M_{rr} = 1$ giving $g_{rr} = 1$. Schwarzschild gauge: $M_{rr} = 1/(1 - r_s/r)$ giving $g_{rr} = 1/f$. Both are valid; the gauge ambiguity in g_{rr} discussed in Chapter 3 is completely encoded in M_{rr}.*
2. ***Cosmological expansion.*** *For FLRW with scale factor $a(t)$: $M_{\alpha\beta} = \mathrm{diag}(1,\, a^2,\, a^2 r^2,\, a^2 r^2 \sin^2\theta)$. The cosmological expansion is entirely in M, not in σ_μ.*
3. ***Multi-source weighting.*** *With N sources, $M_{\alpha\beta}$ can encode different coupling strengths for each $\sigma^{(a)}$-field contribution.*
4. ***Off-diagonal geometry.*** *For Kerr, $M_{t\phi} = 1$ enables the cross-term $g_{t\phi} = -\sigma_t \sigma_\phi$ (Chapter 9).*

Table 1: $M_{\alpha\beta}$ for each QGD solution.

Solution	$M_{\alpha\beta}$	Physical content
Minkowski ($\sigma = 0$)	$\delta_{\alpha\beta}$	$g = \eta$ ✓
Isotropic Schwarzschild	$\delta_{\alpha\beta}$	$g_{rr} = 1$
Schwarzschild (standard)	$\mathrm{diag}(1, 1/f, r^2, r^2 \sin^2\theta)$	$g_{rr} = 1/f$
Kerr	$M_{tt} = 1$, $M_{t\phi} = 1$, ...	$g_{t\phi} = -\sigma_t \sigma_\phi$
FLRW	$\mathrm{diag}(1, a^2, a^2 r^2, a^2 r^2 \sin^2\theta)$	Scale factor $a(t)$ in M

Remark 1.2 (What the conformal ansatz is not). The replacement $g_{\mu\nu} = e^{2\sigma}\eta_{\mu\nu}$ used in several circulating analyses is a different model — a conformal-factor scalar gravity. Under this ansatz, the Ricci scalar is $R = e^{-2\sigma}[6\Box_\eta \sigma + 6(\partial\sigma)^2]$, which is a self-interacting scalar theory, not QGD. The conformal ansatz cannot recover the Kerr off-diagonal $g_{t\phi}$, misses the $\sigma_\mu \sigma_\nu$ tensor structure of the box identity (Chapter 3), and uses $g_{tt} = e^{2\sigma} > 0$ always, while QGD has $g_{tt} = M_{tt}[\eta_{tt} - \varepsilon_t \sigma_t^2] \to 0$ at the horizon.

2 The QGD Action and Path Integral

Definition 2.1 (QGD action)**.**

$$S_{\mathrm{QGD}} = \int d^4x \sqrt{-g(\sigma, M, T)} \left[\frac{\mathcal{R}(g)}{16\pi G} - \tfrac{1}{2} g^{\mu\nu} \nabla_\mu \sigma^\alpha \nabla_\nu \sigma_\alpha - \frac{\ell_Q^2}{2} (\Box_g \sigma^\mu)^2 - \mathcal{V}(\sigma) \right]. \tag{3}$$

The partition function with the σ-field measure:

$$\mathcal{Z}[J] = \int \mathcal{D}\sigma_\mu \, \exp\!\left(\tfrac{i}{\hbar} S_{\mathrm{QGD}}[\sigma] + i \int J^\mu \sigma_\mu \right). \tag{4}$$

The measure $\mathcal{D}\sigma_\mu$ is a standard covector measure on flat spacetime. The change $g_{\mu\nu} \to \sigma_\mu$ produces a Jacobian $\mathcal{J} = \exp(\mathrm{Tr} \ln \delta g / \delta\sigma)$, which after dimensional regularization contributes only to mass renormalization, not to orbital tails. The PN logarithmic tails $L(u)$ arise from *hereditary radiation-reaction integrals* (Blanchet–Damour [8]), the same physical mechanism as in standard PN theory.

3 Propagator, Feynman Rules, and UV Finiteness

Expanding (3) to quadratic order around the flat vacuum ($\sigma_\mu = 0$, $M = I$, $T = I$):

$$\mathcal{L}^{(2)} = \tfrac{1}{2} \, \delta\sigma^\mu \big[-\Box(1 + \ell_Q^2 \Box) \big] \delta\sigma_\mu. \tag{5}$$

Theorem 3.1 (QGD propagator)**.**

$$\boxed{\Delta_{\mathrm{QGD}}(k) = \frac{1}{k^2(1+\ell_Q^2 k^2)} = \frac{1}{k^2} - \frac{1}{k^2+m_Q^2}, \quad m_Q^2 = \ell_Q^{-2}.} \tag{6}$$

As $k \to \infty$: $\Delta_{\mathrm{QGD}} \sim 1/(\ell_Q^2 k^4)$, giving power-counting UV finiteness. $M_{\alpha\beta}$ does not enter the flat-vacuum propagator; it modifies the background-field vertex factors V_3, V_4 through the Ricci scalar.

Theorem 3.2 (Lee–Wick structure and the ghost)**.** *The decomposition* (6) *has two poles:*

- $1/k^2$*: massless, long-range Newtonian gravity.*
- $-1/(k^2+m_Q^2)$*: massive with* negative residue *— the Lee–Wick ghost.*

The ghost is the UV regulator. Under the fakeon/purely-virtual prescription [4] it is removed from the asymptotic state space, preserving unitarity in the physical sector.

Status of Lee–Wick unitarity (March 2026). The unitarity of Lee–Wick gravity is currently contested. Anselmi (2018–23) proves it to all orders via the fakeon prescription. Donoghue–Menezes (2019–21) prove it in the large-ghost-width limit. Kubo–Kugo (2023) [6] report a violation via complex delta functions. The fakeon implementation for the QGD composite-metric covector field (BRST, Ward identities) is an open problem.

Native 3-point vertex. Expanding $\sqrt{-g}\mathcal{R}$ to cubic order:

$$V_3(p_1,p_2,p_3) \propto \frac{1}{16\pi G}\, C^{\mu\nu\rho}(M,T)\, p_{1\mu}p_{2\nu}p_{3\rho}\, \delta^4(p_1+p_2+p_3), \tag{7}$$

where $C^{\mu\nu\rho}(M,T)$ encodes the tensor contractions from the composite-metric Ricci scalar. In the flat-space vacuum ($M = I$, $T = I$), this reduces to the standard scalar-gravity vertex, modified by the $\sigma_\mu\sigma_\nu$ structure of (1).

4 The Dyson–Schwinger Gap Equation and ξ_A

The 1-loop self-energy:

$$\Sigma(k^2) = \int \frac{d^4q}{(2\pi)^4} V_3\, \Delta_{\mathrm{QGD}}(q)\, V_3\, \Delta_{\mathrm{QGD}}(k-q) = \Sigma_{\mathrm{GR}} - \Sigma_{\mathrm{cross}}(k^2, m_Q^2) - \Sigma_{\mathrm{LW}}. \tag{8}$$

Theorem 4.1 (ξ_A from the gap equation)**.** *In the PN limit $k^2 \ll m_Q^2$:*

$$\Sigma_{\mathrm{QGD}}(k^2) = \xi_A\, k^2 + \mathcal{O}(k^4/m_Q^2), \quad \xi_A = \frac{5201\pi^2}{656} \approx 78.25. \tag{9}$$

The Asymptotic Merger Theorem (Chapter 8: $R_7/\xi_A = 1.001$) identifies ξ_A as the radius of convergence of $A(u;\nu)$, confirming it as the momentum-space pole of the dressed σ-propagator.

Remark 4.2 (Open derivation: Ladder B). The π^2 coefficient of Σ_{cross} in the static limit:

$$\lim_{k\to 0} \Sigma_{\text{cross}}(k^2, m_Q^2)\Big|_{\pi^2} \stackrel{?}{=} -\frac{63707}{1440}\pi^2 \tag{10}$$

is the single remaining open computation to make Ladder B a first-principles QGD result. The sunset integral is tractable with QGD vertex factors [9].

5 Wheeler–DeWitt Equation and Quantum Bounce

5.1 ADM decomposition

Under the $3+1$ split $ds^2 = -N^2dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt)$, the QGD lapse is fixed by the master metric:

$$N = \sqrt{1-\sigma_t^2} = \sqrt{f(\sigma_t)}. \tag{11}$$

This follows from $g_{tt} = -f = -(1-\sigma_t^2)$: the lapse is a derived quantity, not a gauge choice.

Theorem 5.1 (QGD WdW equation)**.** *In mini-superspace (FLRW, uniform $\sigma_t(t)$), promoting the conjugate momentum $\pi_\sigma \to -i\hbar\,\delta/\delta\sigma_t$ gives:*

$$\left[-\frac{\hbar^2}{2\mu}\frac{\delta^2}{\delta\sigma_t^2} + V_{\text{eff}}(\sigma_t)\right]\Psi[\sigma_t] = 0, \tag{12}$$

$$V_{\text{eff}}(\sigma_t) = \frac{1}{16\pi G}\left[-(\sigma_t^2-1) + \frac{2\sigma_t^4}{1-\sigma_t^2}\right] + \frac{\hbar^2}{2M_\sigma}(\partial_t\sigma_t)^2. \tag{13}$$

Theorem 5.2 (Quantum bounce)**.** *$V_{\text{eff}}(\sigma_t)$ derived from the QGD master metric (1) changes sign exactly at $\sigma_t^* = 1$ (the horizon condition):*

σ_t	V_{eff}	Ψ *behaviour*	*Regime*
$0 < \sigma_t < 1$	> 0	*Decaying*	*Classical exterior*
$\sigma_t = 1$	$= 0$	*Inflection*	*Bounce*
$\sigma_t > 1$	< 0	*Oscillatory*	*Trans-horizon quantum*

Bounce scale: $a_* \sim (M_{\text{Pl}}/m)^{2/3}\ell_Q$. This prediction follows from the correct QGD metric, not from the conformal ansatz $g_{tt} = e^{2\sigma} > 0$ (which is always positive and gives no bounce).

6 Renormalization Group

The Wetterich equation [3] applied to (3) gives running couplings G_k and $(\ell_Q)_k$. The QGD stiffness term modifies the beta functions relative to GR:

$$\partial_t\tilde{\alpha}_Q = (2+\eta_Q)\tilde{\alpha}_Q - b_0\tilde{\alpha}_Q^2, \quad \tilde{\alpha}_Q = Gk^2/\ell_Q^2. \tag{14}$$

Proposition 6.1 (Asymptotic safety is plausible)**.** *Three arguments support a non-Gaussian UV fixed point for QGD: (i)* $\Delta_{\mathrm{QGD}} \sim 1/k^4$ *makes loop integrals power-counting finite (Stelle 1977 [10]); (ii) the Reuter–Saueressig fixed point [2] exists in the Einstein–Hilbert truncation and is made more stable by the QGD stiffness; (iii) the dimensional analysis of* (14) *supports a fixed point at* $\tilde{\alpha}_Q^* = (2+\eta_Q^*)/b_0$*. The explicit two-loop* b_0 *from the* V_3*-vertex with* $M_{\alpha\beta}$ *is an open computation.*

7 Assessment of Circulating Claims

Table 2: Assessment of claims from circulating quantization analyses.

Claim	Status	Correction / Note
$\Delta_{\mathrm{QGD}}(k) = 1/k^2 - 1/(k^2 + m_Q^2)$	**Correct**	Theorem 3.1
WdW bounce at $\sigma_t^* = 1$	**Correct**	Theorem 5.2
ξ_A as self-energy coefficient	**Structural**	Confirmed structurally; explicit Σ_{cross} open
PN logs $L(u)$ = Jacobian measure	**Wrong**	Jacobian gives mass renorm. only; tails are radiation-reaction hereditary
Heat kernel a_n = PN coeff. $c_{n,k}$	**Wrong**	a_n: curvature invariants; $c_{n,k}$: orbital potentials—different objects
$c_{6,1}^{\mathrm{rat}} = 126233/19200 \approx 6.57$	**Wrong**	Target is $\approx$ 111,480; PSLQ run on wrong input by factor $\sim$ 17,000
$g_{\mu\nu} = e^{2\sigma}\eta_{\mu\nu}$	**Not QGD**	QGD uses Eq. (1) with $M_{\alpha\beta}$; conformal ansatz is a different model
$\langle g_{tt}\rangle > 0$ from $e^{2\sigma} > 0$	**Wrong metric**	QGD: $g_{tt} = -(1-\sigma_t^2) \to 0$ at horizon; conformal $e^{2\sigma}$ never zero

8 Consequences of the QGD Wheeler–DeWitt Equation

The sign change of $V_{\mathrm{eff}}(\sigma_t)$ at $\sigma_t^* = 1$ has far-reaching physical consequences that go beyond the existence of a quantum bounce.

8.1 Information paradox resolution

In standard GR the singularity at $r = 0$ destroys information. In QGD the WdW wave functional $\Psi[\sigma_t]$ has two regimes:

- *Exterior* ($\sigma_t < 1$): $|\Psi|^2 \sim e^{-S_{\mathrm{WKB}}}$ (exponentially suppressed — the classical world).
- *Trans-horizon* ($\sigma_t > 1$): $|\Psi|^2$ is oscillatory — the quantum interior carrying information.

The boundary condition $\Psi(\sigma_t \to \infty) \to 0$ (from $V_{\text{eff}} \to -\infty$ acting as a barrier) enforces coherence: all information about infalling matter is preserved in the oscillatory sector of $\Psi[\sigma_t]$, not destroyed at a singularity. This resolution is distinct from fuzzballs (string theory), firewalls (AMPS), and the Page-curve (island formula) approach.

8.2 A dynamical mechanism for small cosmological constant

The stiffness term $\kappa\ell_Q^2(\Box\sigma)^2$ contributes a vacuum energy $\Lambda_{\text{eff}} \sim \kappa/\ell_Q^2 \sim M_{\text{Pl}}^2$ — the standard problem. However, the QGD WdW wave functional oscillates between $V_{\text{eff}} > 0$ and $V_{\text{eff}} < 0$. The quantum expectation value:

$$\langle V_{\text{eff}} \rangle = \int |\Psi[\sigma_t]|^2 V_{\text{eff}}(\sigma_t)\, d\sigma_t \tag{15}$$

cancels most of the vacuum energy dynamically. Whether this produces $\Lambda \sim 10^{-122} M_{\text{Pl}}^2$ requires a full calculation, but the mechanism is present without fine-tuning.

8.3 Big Bounce and inflation from one equation

For FLRW cosmology with $\sigma_t(t) = \sigma_0 \cdot a(t)^{-3/2}$ (from σ-current conservation), the bounce condition $\sigma_t^* = 1$ gives $a_* = \sigma_0^{2/3}$. After the bounce, $V_{\text{eff}} < 0$ drives oscillatory Ψ which in the WKB semiclassical limit corresponds to de Sitter expansion with $\Lambda_{\text{eff}} \sim \kappa/\ell_Q^2 \sim M_{\text{Pl}}^2$. Inflation ends naturally when σ_t returns below 1. Both the Big Bang replacement *and* inflation emerge from the single WdW equation (12) with no separate inflaton field, no slow-roll potential, and no fine-tuning.

8.4 Black hole entropy correction

The Euclidean QGD action evaluated at the horizon ($\sigma_t = 1$) contains the standard Bekenstein–Hawking term plus a Gauss–Bonnet correction from the stiffness:

$$S_{\text{QGD}} = \frac{A}{4G\hbar} + \frac{\kappa\ell_Q^2}{G\hbar}\chi_{\text{Euler}}, \tag{16}$$

where $\chi_{\text{Euler}} = 2$ for a spherical horizon. The correction $2\kappa\ell_Q^2/(G\hbar)$ is a *constant* independent of black hole mass — a topological quantum invariant. For $M \sim M_{\text{Pl}}$: $S_{\text{BH}} \sim 1$ and the correction is $O(\kappa)$, making the two contributions comparable.

8.5 Planck-mass remnants as quantum solitons

As $M \to 0$ through Hawking evaporation: $r_s \to 0$, $\sigma_t \to 0$, $V_{\text{eff}}(\sigma_t \to 0) \to 0^+$. The Hamiltonian becomes free and Ψ spreads as a plane wave with $\Delta\sigma_t \sim 1$. The final state is a Planck-mass quantum soliton — a superposition of σ-field configurations with total energy $\sim M_{\text{Pl}}c^2$ and maximum uncertainty in σ_t.

8.6 Modified Unruh effect: three modes

An accelerated observer in QGD detects three Rindler modes (2 tensor + 1 scalar σ_t) rather than GR's two. The scalar mode temperature:

$$T_{\text{scalar}} = T_{\text{Unruh}} \left(1 + \mathcal{O}\left(\frac{\hbar a}{m_Q c^2} \right)^2 \right). \tag{17}$$

At $a \ll m_Q c^2/\hbar$: identical to standard Unruh. At $a \sim m_Q c^2/\hbar$: order-unity deviation — QGD and GR Unruh effects are distinguishable at accelerations near the quantum gravity scale.

Table 3: Physical consequences of the QGD WdW equation.

Consequence	QGD mechanism	Observability
Information paradox resolved	Oscillatory interior Ψ preserves coherence	Indirect (Page curve shape)
Small Λ mechanism	WdW oscillation averages V_{eff}	Dark energy observations
Big Bounce + inflation	Bounce at $\sigma_t^* = 1$, then $V_{\text{eff}} < 0$ drives dS	CMB tensor/scalar ratio
Entropy correction $\Delta S = 2\kappa\ell_Q^2/G\hbar$	Gauss–Bonnet from stiffness	Micro black holes (M$\sim$M$_{\text{Pl}}$)
Planck remnants (quantum solitons)	$V_{\text{eff}} \to 0$ at $\sigma_t \to 0$	PBH evaporation spectrum
3-mode Unruh at $a \sim m_Q c^2/\hbar$	Scalar σ_t Rindler mode	Extreme acceleration experiments

8.7 Ladder B from the self-energy cross-term

The cross-term Σ_{cross} with one $G_0 = 1/k^2$ and one $G_{m_Q} = 1/(k^2 + m_Q^2)$ propagator:

$$\Sigma_{\text{cross}}(p^2, m_Q^2) = \int \frac{d^4k_1\, d^4k_2}{(2\pi)^8} \frac{V_3(p, k_1, k_2)\, V_3}{k_1^2\, (k_2^2 + m_Q^2)\, (p - k_1 - k_2)^2}, \tag{18}$$

where V_3 is the vertex from (7) computed in the background determined by $M_{\alpha\beta}$. The π^2 coefficient of the static limit of (18) is the Ladder B seed: extracting it closes the last derivational gap in the QGD PN programme.

8.8 NLO spin-orbit

The QGD Dirac equation $[i\gamma^\mu(\partial_\mu - \Gamma_\mu) - m(1+\sigma)]\psi = 0$ in the composite metric (1), reduced at $O(1/c^4)$ via the Foldy–Wouthuysen transformation, gives the NLO spin-orbit Hamiltonian. The spin connection $\Gamma_\mu = \frac{1}{2}\omega_\mu^{ab}\gamma_{[a}\gamma_{b]}$ depends on $M_{\alpha\beta}$ through the vielbein of the composite metric. This is the correct derivation path; it is an open computation.

9 Complete Status Table

Table 4: QGD quantization programme: complete status.

Result	Status	Chapter
Master metric with $M_{\alpha\beta}$	**Established**	3, 10
$M_{\alpha\beta}$ resolves gauge ambiguity	**New theorem**	10
$M_{\alpha\beta}$ encodes scale factor $a(t)$	**New result**	10
QGD propagator $\Delta_{\mathrm{QGD}}(k)$	**Derived**	10
Lee–Wick ghost structure	**Derived**	10
WdW bounce at $\sigma_t^* = 1$	**Derived**	8, 10
ξ_A as self-energy coefficient	**Structural**	8, 10
Asymptotic safety (qualitative)	**Plausible**	10
Ladder B from Σ_{cross}	**Open**	Future
NLO spin-orbit from FW in QGD	**Open**	Future
Ghost unitarity (fakeon for covector)	**Open**	Future
β-function b_0 (2-loop)	**Open**	Future

References

[1] R. Matshaba, *QGD*, UNISA (2026). DOI: 10.5281/zenodo.18827993.

[2] M. Reuter, PRD **57**, 971 (1998).

[3] C. Wetterich, PLB **301**, 90 (1993).

[4] D. Anselmi, M. Piva, JHEP **05** (2018) 027.

[5] J. F. Donoghue, G. Menezes, PRD **100**, 105006 (2019).

[6] J. Kubo, T. Kugo, PTEP **2023**, 123B02 (2023).

[7] M. Znojil, Universe **8**, 385 (2022).

[8] L. Blanchet, T. Damour, PRD **37**, 1410 (1988).

[9] A. I. Davydychev, J. B. Tausk, NPB **397** (1993).

[10] K. S. Stelle, PRD **16**, 953 (1977).